Bilal Jellaoui

Data Scientist · Data Analyst · Machine Learning Engineer · GENERATIVE AI

Master's student (Year 2) in Big Data and AI— **Seeking PFE internship · February 2027 · 6 months**

Kénitra, Morocco +212 6 87 42 13 17 bilaljellaoui@gmail.com

linkedin.com/in/bilal-jellaoui github.com/Bilal-Jellaoui Portfolio

PROFILE

Master's student in Big Data, Artificial Intelligence and Advanced Applications at Ibn Tofail University, Kénitra. Academic projects in generative AI, NLP and machine learning, from data preparation to integrating models into Flask/REST API applications. Looking for a six-month end-of-studies internship (PFE) in AI & innovation starting February 2027.

ACADEMIC PROJECTS

Multilingual AI chatbot for psychological support GitHub

Python · Flask · DistilBERT · Llama 3.3 · Groq API · SQLite

- Developed a multilingual chatbot (Arabic/Darija, French, English): 105 intents, 12 emotions; F1 scores of 93.4% and 90.9%.
- Integrated Llama 3.3, crisis-situation detection before replying, conversational memory and REST API.

Bank fraud detection GitHub

Python · scikit-learn · XGBoost · SMOTE · Flask · REST API

- Analysis of 284,807 transactions (0.173% fraud); model comparison and tuning with cross-validation.
- XGBoost: ROC-AUC 0.982 and F1 0.871; Flask application with REST API, SQLite logging and dashboard.

Customer segmentation and customer value GitHub

Data Mining · RFM segmentation · Logistic regression · K-means

- Preprocessed 541,909 Online Retail transactions and built an RFM table for 4,338 customers.
- Logistic regression: AUC 0.911 and accuracy 83.6%; K-means: 4 segments to guide marketing decisions.

Real-time web log streaming — team lead (6 members) GitHub

Java 17 · Spring Boot · Apache Kafka · Kafka Streams · Docker · GitHub Actions

- Designed the log parser and the Kafka Streams engine in a real-time pipeline compared against a Java engine.
- 3-broker Kafka cluster on Docker, JUnit tests, continuous integration and alerts on HTTP 5xx errors.

ZooBeco — animal image classification GitHub

Python · CNN · Kivy · Image classification

- Classification of 6 animal classes with 89.67% accuracy and an "Unknown" class below 70% confidence.
- Mobile app with camera, gallery and confidence score display.

TECHNICAL SKILLS

Programming and data: Python, Java, C, C++, SQL, Shell/Linux, pandas, NumPy, EDA, data cleaning, feature engineering, algorithm optimization

AI and Machine Learning: scikit-learn, supervised/unsupervised learning, Deep Learning, CNN, Transformers, NLP, LLM (Llama 3.3), DistilBERT, TensorFlow, PyTorch, spaCy, Groq API, Flask, REST API

Big Data and BI: Hadoop, HDFS, YARN, HBase, Hive, Kafka, Kafka Streams, data pipelines, Power BI, KPI, Matplotlib, Seaborn

Tools: Git, GitHub, Docker, GitHub Actions, Linux, Jupyter Notebook, VS Code, Hugging Face Transformers, operating systems, computer architecture

EDUCATION

Master's in Big Data, AI and Advanced Applications 2025–2027

Ibn Tofail University · Faculty of Sciences, Kénitra

Bachelor's in Mathematics and Computer Science (SMI) 2025

Ibn Tofail University · Kénitra

Baccalaureate in Mathematical Sciences A 2021

Lycée Mohammed V · Sidi Kacem

STRENGTHS AND LANGUAGES

Analytical mind · Problem solving · Scientific rigor · Communication · Autonomy · Teamwork

Arabic: native | **French:** good | **English:** intermediate